

SOC-5811 Week 8: Introduction to causal inference

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MULTIPLE PREDICTORS

- ▶ The coefficient β_k is the average or expected difference in outcome y , comparing two people who differ by one unit in the predictor x_k while being equal in all the other predictors. This is sometimes stated in shorthand as comparing two people (or, more generally, two observational units) that differ in x_k with all the other predictors held constant.

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- ▶ A one unit increase in X is associated with a β_x increase in Y , all else constant.

MULTIPLE PREDICTORS

$$y_i = X_i + Z_i$$

I want to estimate the effect of X on Y conditional on Z and I believe the effect of X does not *depend* on the level of Z . These effects are *independent* of one another.



MULTIPLE PREDICTORS

$$y_i = X_i + Z_i + (X_i \times Z_i)$$

I want to estimate the effect of X on Y conditional on Z , but I also believe the effect of X *depends* on the level of Z .



MULTIPLE PREDICTORS

- What kind of hypotheses do interaction terms let me examine? What does an estimate of 0 on an interaction term mean?



MULTIPLE PREDICTORS

- ▶ What kind of hypotheses do interaction terms let me examine? What does an estimate of 0 on an interaction term mean?
- ▶ **Interaction coefficient:** Tests whether the effect of X is different across levels of Z .



REVIEW

Research Question: What is the average association between income and commute time across the twin cities, controlling for city?

- ▶ Y = commute time (continuous)
- ▶ X = income (continuous)
- ▶ Z = city (categorical: Minneapolis, St Paul)

A: $y_i = \beta_0 + \boxed{\beta_1}(X)$

B: $y_i = \beta_0 + \boxed{\beta_1}(X) + \beta_2(Z)$

C: $y_i = \beta_0 + \beta_1(X) + \boxed{\beta_2}(Z)$

D: $y_i = \beta_0 + \beta_1(X) + \beta_2(Z) + \boxed{\beta_3}(X \times Z)$

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Research Question: Is the average association between income and commute time different depending on the city?

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What is the null hypothesis here?

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$$y_i = \beta_0 + \beta_1(\text{income}) + \beta_2(\text{city}=\text{M}) + \boxed{\beta_3}(\text{income} \times \text{city}=\text{SP})$$



$$y_i = 10 + 3(\text{income}) + 20(\text{city}=\text{M}) + \boxed{0}(\text{income} \times \text{city}=\text{SP})$$

- What is the effect of income on commute time in Minneapolis?

$$y_i = 10 + 3(\text{income}) + 20(\text{city}=\text{M}) + \boxed{0}(\text{income} \times \text{city}=\text{SP})$$

- ▶ What is the effect of income on commute time in Minneapolis?
- ▶ What is the effect of income on commute time in St Paul?

$$y_i = 10 + 3(\text{income}) + 20(\text{city}=\text{M}) + \boxed{0}(\text{income} \times \text{city}=\text{SP})$$

- ▶ What is the effect of income on commute time in Minneapolis?
- ▶ What is the effect of income on commute time in St Paul?
- ▶ What is the difference in the effect of income on commute time between St Paul and Minneapolis?

CAUSAL INFERENCE

- Transitioning to causal inference: How do I know if X has a *causal effect* on Y ?

CONFOUNDING

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- ▶ People who drink more wine are less likely to die ($p < 0.05$)!



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- ▶ People who drink more wine are less likely to die ($p < 0.05$)!
- ▶ **Interpretation:** People who drank more wine are less likely to die *because* they drank more wine; *if* people drink more wine, *then* they will be less likely to die.

CONFOUNDING

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How might I get a coefficient estimate that is closer to this interpretation?



CONFOUNDING

$$\text{mortality rate} = \beta_0 + \beta_1 \text{wine} + \beta_2 \text{age} + \beta_3 \text{income} + \beta_4 \text{education} + \beta_5 \text{social}$$

- ▶ People who drink more wine are less likely to die comparing within levels of age, income, education, and social networks ($p < 0.05$)!

CONFOUNDING

$$\text{mortality rate} = \beta_0 + \beta_1 \text{wine}$$

How might we avoid having to control for so many things? In other words, how might we guarantee the *only* difference across our comparisons of drinking is the drinking?



DAGs

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- ▶ **Nodes:** random variables.
- ▶ **Arrows:** causal effects between two nodes that move in the direction of the arrow.

Wine $\longrightarrow$ Mortality

Ground rules:

- ▶ Causality runs in one direction: forward through time.

DAGs

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- ▶ DAGs represent chains of events.
- ▶ DAGs operate in counterfactuals: use use them to compare hypothetical worlds, scenarios, interventions.

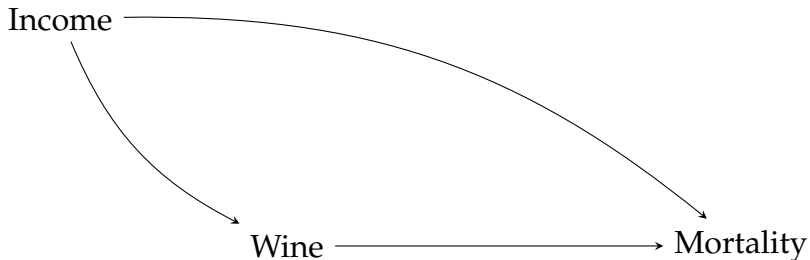
Why DAGs?

- ▶ Force you to think carefully about the logic of how different variables are related to each other.

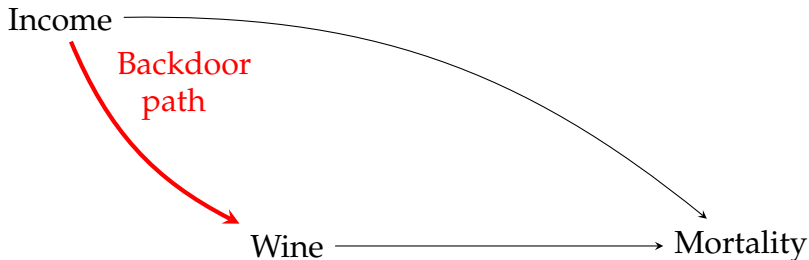
Why DAGs?

- ▶ Force you to think carefully about the logic of how different variables are related to each other.
- ▶ Writing out a DAG prior to your analysis highlights the necessity of bringing in substantive, or domain, knowledge into your study.

DAGs



DAGs



CONFOUNDING

In this case, if we are interested in the causal effect of wine on mortality, we would say that **income** acts as a **confounder**. Because income may have an effect on both wine drinking and mortality risk, the effect of wine estimated with a model not controlling for income would be at least partially spurious.

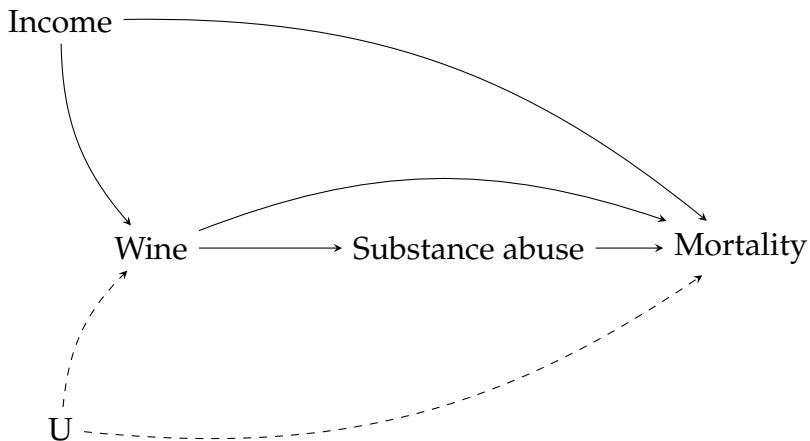
$$\text{mortality rate} = \beta_0 + \beta_1 \text{wine} + \beta_2 \text{income}$$



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- ▶ Mediators, mechanisms, etc.

DAGs



REVIEW

- ▶ Return on Wednesday.



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Research Question: What is the difference in grandchild wealth by grandparent race, controlling for grandparent wealth?

- ▶ Y = grandchild wealth (continuous)
- ▶ X = grandparent wealth (continuous)
- ▶ Z = grandparent race (categorical: Black, White)

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Research Question: What is the difference in the average association between grandparent wealth and grandchild wealth by race?

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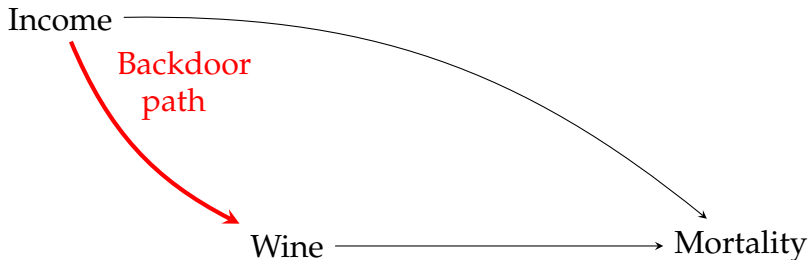
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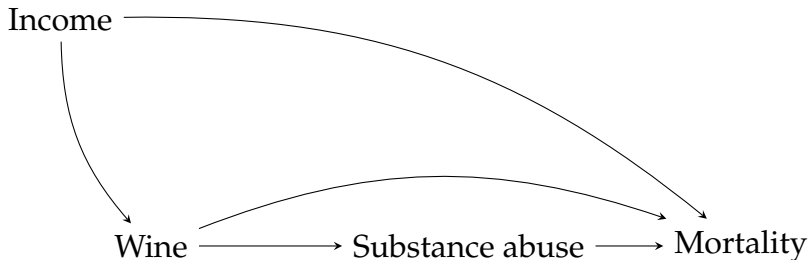
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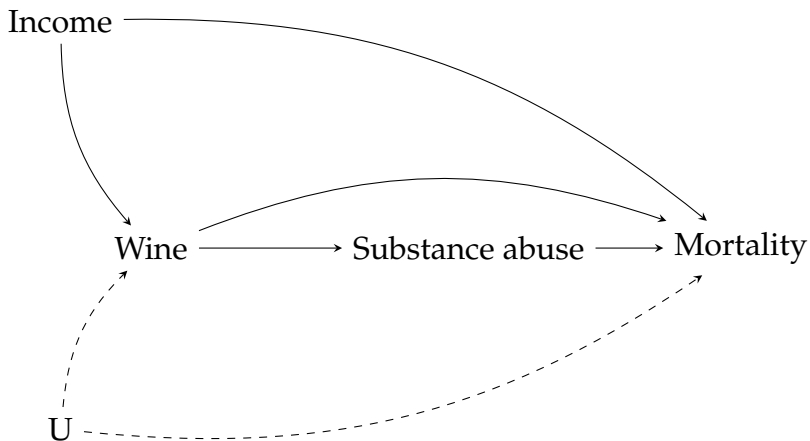
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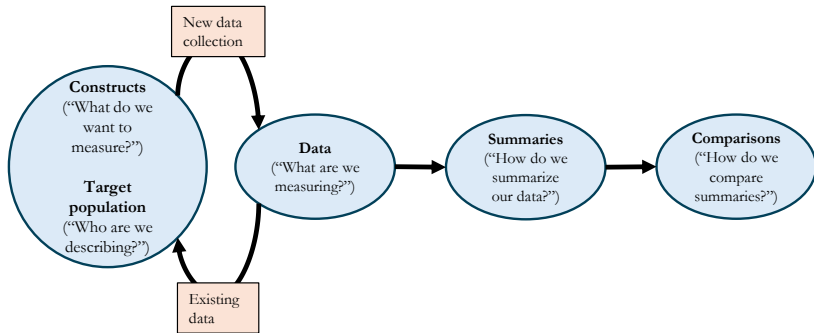


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DAGs





REVIEW

1. Choose a functional form for the model.
2. Choose an estimator (i.e., objective function) that relates the model to real observed data (e.g., OLS).
3. Fit the model (e.g., estimate parameters) by optimizing the objective function.



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$$y_i = \beta_0 + \boxed{\beta_1} x_i + \epsilon_i$$

2. Choose an estimator (i.e., objective function) that relates the model to real observed data (e.g., OLS).

$$\text{Sum of squared residuals} = \sum_{i=1}^n (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2$$

3. Fit the model (e.g., estimate parameters) by optimizing the objective function.

REVIEW

1. Choose a functional form for the model.

$$y_i = \beta_0 + \boxed{\beta_1} x_i + \epsilon_i$$

2. We already know the best estimator for the linear model: OLS.
3. We already have pre-built R functions to fit the estimator to data.



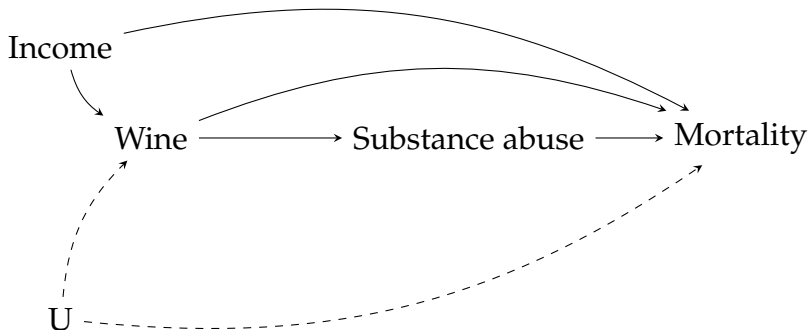
REVIEW

1. We don't know the functional form for the model; we must justify our target comparison, or estimand, using theory.

$$y_i = \beta_0 + \boxed{\beta_1} x_i + \epsilon_i$$

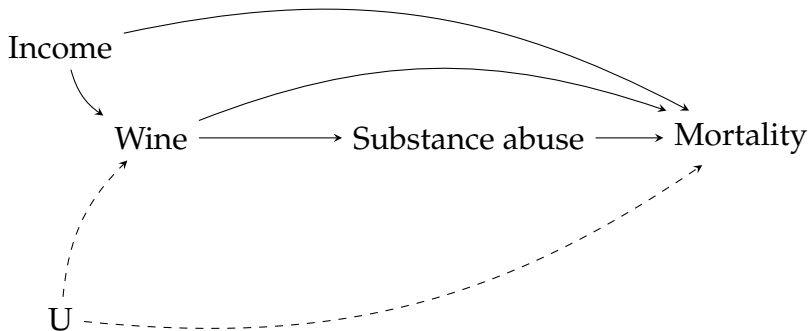
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DAGs

$$\text{Mortality}_i = \beta_0 + \boxed{\beta_1} \text{Wine}_i + \beta_2 \text{Income} + \epsilon_i$$



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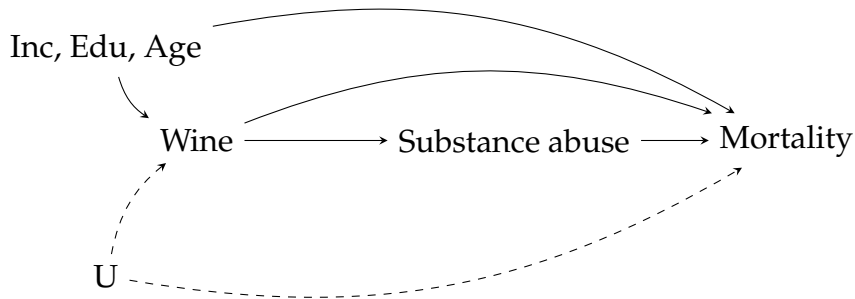


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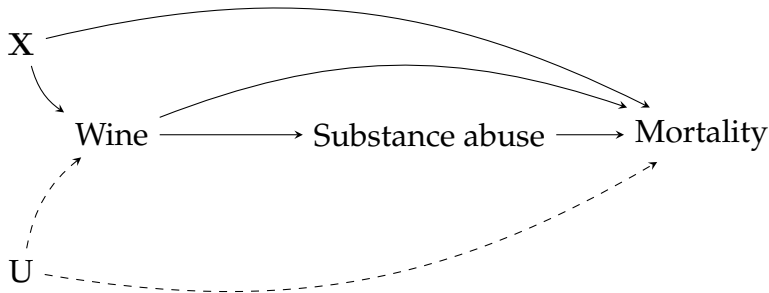
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- ▶ **Irrelevance:** Remove variables that are not on the causal pathway between your key treatment and outcome.

DAGs



DAGs

$$\text{Mortality}_i = \beta_0 + \boxed{\beta_1} \text{Wine}_i + \beta_2 \mathbf{X} + \epsilon_i$$



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- ▶ What they are really saying: “The comparison in which you are not controlling for X is not useful; I’m assuming you want to make the comparison where you control for X.” This can be an argument on theoretical grounds about the utility of various research questions.
- ▶ There may be very good reasons to control for X! But it *completely* depends on your research question; more specifically, exactly what **comparison** you say you’re trying to make.



CONCEPTUAL MODELS

What is the difference between a DAG and a conceptual model?



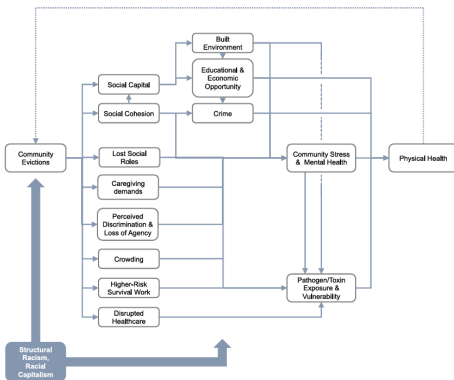


Fig. 1. Theoretical model: evictions as a community health exposure

Note: This diagram is best read from left to right, starting with community evictions and ending with physical health. Here, structural racism and racial capitalism (A) shape who is at risk of experiencing community evictions as well as (B) modify the relation between community evictions and our theorized mediators, as well as between these mediators and health.

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- ▶ **(3)** Refine conceptual model into a DAG (this may include drawing the “causal field”).
- ▶ **(4)** Think systematically and explicitly about the potential effect of bias.

EVICTED EXAMPLE

On the board.



DAGs

Let's workshop a DAG on the board for someone's research project.

